Policy-Grounded Agentic AI for Claim Readiness and Appeal Support

ABSTRACT

Insurance claim processing has become increasingly imbalanced between insurers and policyholders. When a claim is delayed, queried, or denied, insurers use automated systems to adjudicate the claim within seconds, while the policyholder receives a notice written in technical language with a short response window, often 15–30 days. Policyholders often struggle to interpret denial notices, assess whether an appeal is worth pursuing, and identify the evidence needed to support their case. As a result, many claims go unappealed or are appealed without adequate policy-grounded justification, and avoidable rejections occur due to incomplete or inconsistent documentation at the time of filing.

Existing solutions mainly rely on insurance agents, customer support, or generic AI chatbots, which offer only generic guidance and cannot reason over a policyholder's own policy document. Recent research has explored LLMs, Retrieval-Augmented Generation (RAG), and Multi-Agent AI for policy understanding, claim analysis, and appeal generation, but these approaches typically address individual tasks in isolation. Notably, the majority of existing work concentrates on claim verification — validating documentation and eligibility at the time of filing — while very few, if any, solutions focus specifically on denial-appeal support, leaving policyholders without dedicated tools once a claim has already been rejected.

To address this gap, this project proposes a Multi-Agent Agentic AI system that assists policyholders throughout the insurance claim lifecycle. Its denial-appeal engine re-examines a rejected claim against the policyholder's own policy, assesses whether an appeal is worth pursuing, identifies missing evidence, and drafts a formal appeal backed by verified policy clauses. A claim-readiness component flags missing or inconsistent documentation before filing, while a conversational assistant lets policyholders query their policy in natural language with clause-grounded answers. The system is built as a multi-agent pipeline, where specialized agents — for document understanding, policy auditing, evidence verification, and appeal drafting — collaborate to carry a claim from intake to appeal. Document agents extract and OCR raw files, a retrieval agent grounds reasoning in clause-level policy context via RAG, an audit agent checks the denial against policy terms, and a drafting agent composes the appeal. Every cited clause is verified against the source policy to prevent fabrication, and a Human-in-the-Loop checkpoint ensures the policyholder reviews and approves each appeal before submission — enabling more confident, evidence-backed claim decisions.

REFERENCES

[1] Fink, M., Pua, B. B., Hines, J., Brown, K. T., & Solomon, S. B. (2026). *Decreasing Administrative Effort Related to Non-Approval of Image-guided Procedures Using Large Language Models – The DENIED-AI Pilot Study*. Academic Radiology.
https://doi.org/10.1016/j.acra.2026.04.021

[2] *InsuranceRAG: A Multi-Module Retrieval-Augmented AI System for Intelligent Insurance Assistance*. (2026). *Software Impacts*, Volume 28, 100844. Elsevier.
https://www.sciencedirect.com/science/article/pii/S2665963826000345

Supervisor Details:

A.Sangeeta

Assistant Professor

Student Details

Pragallapati Saketh–160123733129

Saketh Vemulapati – 160123733136